Sofya Dymchenko, Ph.D.

Grenoble, France | sofya.dymchenko@gmail.com | svdcvt.github.io

Summary. Applied Scientist with PhD in Computer Science, specialising in deep learning for scientific computing, data-centric machine learning, and high-performance computing (HPC), with publications at NeurIPS. Experience in engineering research software, designing and conducting extensive experiments, supervising small teams, and communicating with technical and non-technical audiences.

Work Experience

Doctoral Researcher, Grenoble, France — Mar 2022 – Apr 2026
Datamove team, INRIA and University of Grenoble-Alpes (UGA)

- Pioneered a research line on query-synthetic active learning for deep PDE surrogates; initiated and led the development of the active sampling branch of Melissa, an open-source HPC/AI framework.
- Designed and implemented Breed [2, 3] for on-the-fly simulation data curation; achieved up to 10x lower surrogate errors under fixed compute budgets; orchestrated GPU/CPU experiments on Leonardo supercomputer.
- Extended this approach with generative modelling: conceived and co-developed scalable diffusion-based OGAS method [1] applied to conditional transformer-based surrogates and neural operators on 2D PDEs.
- Supervised two engineers and two MSc students on Melissa R&D, including work on online pretraining of foundation models for PDE surrogates.

Research Intern, Moscow, Russia — Jun 2021 – Dec 2021
Computational Intelligence Lab, Skolkovo Institute of Science and Technology (Skoltech)

- Extended master's thesis work on optimal transport to computational geometry and computer vision.

Research Intern, Moscow, Russia — Feb 2020 – Aug 2020
Media Algorithms Laboratory, Huawei R&D Center

- Trained generative deep learning models with attention for personalized noise reduction in audio signals.

Education

PhD in Computer Science, Grenoble, France — Mar 2022 – Apr 2026
Laboratory of Informatics of Grenoble (LIG), UGA

- Thesis: High Performance Online Deep Neural Network Training from Synthetic Data with Active Learning for Scientific Computing. The manuscript is rated "Outstanding" (top 5%) by an external reviewer [link].

MSc in Applied Mathematics and Computer Science, Moscow, Russia — Sep 2019 – Jun 2021
Statistical Learning Theory program, Skoltech jointly with HSE; with honors, GPA: 4.89/5

- Implemented a modified optimal transport algorithm, with complexity reduced from quadratic to log-linear, and its application to computational mesh generation [link].
- Managed team delivery of winning data science solutions for marketing applications, from initial concept to final presentation, in industry-supported university hackathons.

BSc in Mathematics, Moscow, Russia — Sep 2015 – Jun 2019
National Research University Higher School of Economics (HSE), GPA: 7.87/10; Minor in Data Science

Additional Experience & Skills

Technical skills: Python (PyTorch, JAX, Matplotlib, Pandas, Seaborn, Scikit-Learn, SciPy, NumPy, WandB), Git, Bash, SLURM, Linux/HPC environments, LaTeX, C++ (basic).

Languages: English (fluent), French (intermediate), Russian (native).

Teaching (2019–2024): led Python course (3 years, 100+ students) incl. curriculum, TA supervision, lectures, won a "Best Course" award (2020); taught Data Science/ML seminars to non-CS students (HSE, UGA).

Academic service: on-site organisation and outreach for GAP workshops [2024][2026]; reviewer for WANT workshops at NeurIPS'23, ICML'24; member of UGA doctoral school student council and LIG laboratory synergy.

Publications

[1] P. Cesar, **S. Dymchenko**, A. Purandare, and B. Raffin. Learning Where to Simulate: Generative Active Sampling for Online PDE Surrogate Training. To appear in *NeurIPS'2026* [preprint]

[2] **S. Dymchenko**, A. Purandare, and B. Raffin. MelissaDL x Breed: Towards Data-Efficient On-line Supervised Training of Multi-parametric Surrogates with Active Learning. In *AI4Science Workshop at SC'2024* [paper] [video]

[3] **S. Dymchenko** and B. Raffin. Loss-Driven Sampling within Hard-to-Learn Areas for Simulation-Based Neural Network Training. In *Machine Learning and the Physical Sciences Workshop at NeurIPS'2023* [paper]